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# vault_backend.py
import hvac
import base64
from typing import Dict, Any

class VaultBackend:
    def __init__(self, url="http://127.0.0.1:8200", token=None):
        self.client = hvac.Client(url=url, token=token)
        if not self.client.is_authenticated():
            raise RuntimeError("Vault authentication failed")

    # --- Transit Engine (encryption / decryption) ---
    def encrypt(self, plaintext: bytes, key_name="rca-master") -> str:
        resp = self.client.secrets.transit.encrypt_data(
            name=key_name,
            plaintext=base64.b64encode(plaintext).decode("utf-8"),
        )
        return resp["data"]["ciphertext"]

    def decrypt(self, ciphertext: str, key_name="rca-master") -> bytes:
        resp = self.client.secrets.transit.decrypt_data(
            name=key_name,
            ciphertext=ciphertext,
        )
        return base64.b64decode(resp["data"]["plaintext"])

    # --- KV Engine (store / read secrets, policies) ---
    def write_secret(self, path: str, data: Dict[str, Any]):
        return self.client.secrets.kv.v2.create_or_update_secret(
            path=path, secret=data
        )

    def read_secret(self, path: str) -> Dict[str, Any]:
        result = self.client.secrets.kv.v2.read_secret_version(path=path)
        return result["data"]["data"]

    def delete_secret(self, path: str):
        self.client.secrets.kv.v2.delete_metadata_and_all_versions(path=path)
        return {"status": "DELETED"}

```